

9. How to Choose a TV

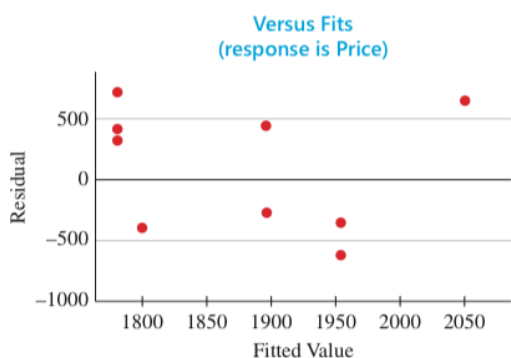
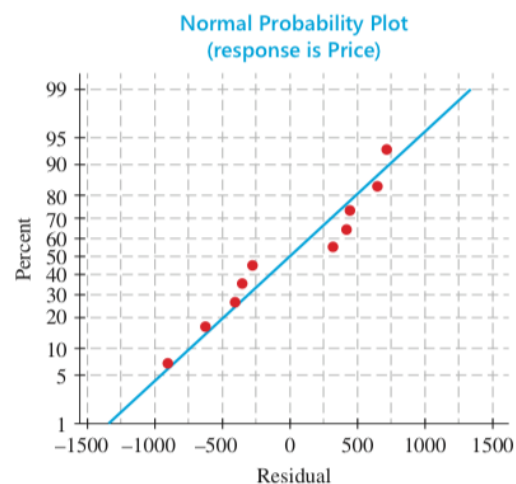
Consumer Reports¹¹ gave the prices and screen sizes for the top 10 LCD TVs in the 46-inch and higher categories.

Does the price of an LCD TV depend on the size of the screen?

Brand	Price (\$)	Size
Sony Bravia KDL-52NX800	2340	52
Samsung LN55C650	1600	55

Brand	Price (\$)	Size
Vizio VF550M	1330	55
Sony Bravia KDL-60EX700	2700	60
Sharp Aquos LED LC-52LE700UN	1620	52
Sony Bravia KDL-46XBR10	2500	46
Samsung UN46C8000	2200	46
Vizio SV472XVT	1400	47
Samsung UN46C7000	2100	46
LG 47LD450	900	47

- a. Assuming that the relationship between size and price is linear, we perform a linear regression, resulting in a value of $r^2 = .027$. What does the value of r^2 tell you about the strength of the relationship between price and screen size?
- b. The diagnostic plots for this data are shown below. Does it appear that either the normality or equal variance assumptions have been violated?



- c. Use a scatterplot to plot price versus screen size for the 10 LCD TVs. Based on the information in part a, which assumption for the linear regression model has been violated?

a. $r^2 = 0.027$ means

model 可解釋的資料變異比例只有 2.7% $\Rightarrow$ price & screen don't have strong relationship.

b. normality

The points roughly follow a straight line

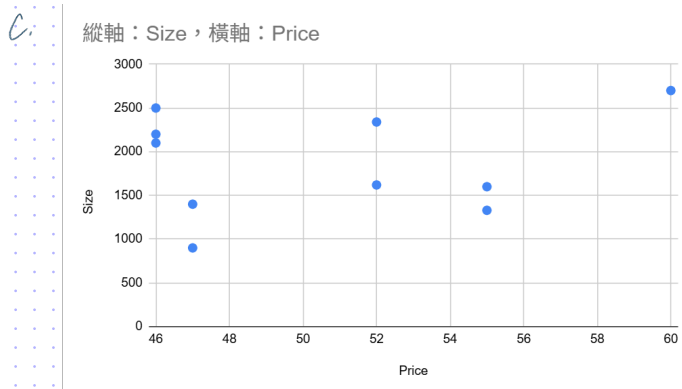
$\Rightarrow$ seems not violated.

equal variance

未看到明顯 pattern & 平均分佈

在 0 上下

$\Rightarrow$ seems not violated.



there's no clearly linear trend.

$\Rightarrow$ linear assumption is violated.